

Amandine PEPIOT

PhD in biomathematics and applied mathematics engineer

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Research

Postdoctoral researcher at **IAME** (UMR1137), Paris, France

2025-...

Evaluation of the effectiveness and cost-effectiveness of prevention and screening strategies for bacterial sexually transmitted infections among men who have sex with men in France.

Doctoral studies

2019-2024

Under the supervision of Romulus Breban (Institut Pasteur)

Title of the thesis: *Eliminating HIV with voluntary testing? A game theoretic perspective* [[thesis](#)]

Lab: IPLESP, UMR-S 1136 Inserm-Sorbonne Université, France

Funding: Sidaction and teaching position at INSA Toulouse

Doctoral school: Pierre Louis de santé publique (ED 393)

Skills: epidemic models (4-disease compartmental model), numerical analysis, Matlab, utility theory, numerical optimization, high-performance computing, scientific paper writing.

Research engineer at **IPLESP** (UMRS1136), Paris, France

2018-2019

- Estimation of incidence, hidden epidemic and time between infection and diagnosis of HIV based on departmental data on new diagnoses. **Skills:** back-calculation model, R, C++, Matlab.

- Analysis of the feasibility of using statistical models to obtain estimates of epidemiological indicators of HIV infection in West Africa (Togo, Burkina Faso, Mali). **Skills:** R, SAS.

Intern at **IPLESP**, Paris, France

2018

Subject: *Evaluating the impact of hometests on the HIV epidemic of men who have sex with men in France* [[report](#)]

Supervisors: Virginie Supervie (Inserm) and Romulus Breban (Institut Pasteur)

Skills: epidemic models, numerical analysis, Matlab

Intern at **Terres Inovia**, Grignon, France

2017

Subject: *Analysis and implementation of linear models to predict the average yield of rapeseed in France*

Supervisor: Sébastien Gervois (Terres Inovia)

Skills: time series, R, Shiny (RStudio)

Teaching

Teaching assistant at INSA Toulouse, France

2022-2024

Theoretical and applied mathematics for 1st, 2nd and 3rd years of engineering study. *Full time, 2 years.*

Teaching mission as a PhD student, Sorbonne Université, Paris, France

2019-2021

Tutorials of biostatistics for first- and second-year medical students. *64 hrs per year.*

Education

Doctoral studies

2019-2024

Sorbonne Université, Paris, France

Degree obtained in December 2024.

Academic exchange - Winter semester

2017-2018

Technische Universität Dresden, Germany

Theory and applications of partial differential equations, finite elements, mathematical applications in biology and risk modeling.

Engineering school - applied mathematics

2015-2018

National Institute of Science and Technology (INSA), Rennes, France

Engineering program focused on theoretical and applied mathematics.

Degree obtained in June 2018.

Classe préparatoire aux grandes écoles MPSI-MP

2013-2015

Lycée Victor Hugo, Besançon, France

*Intensive academic programs in mathematics, physics and industrial sciences designed to prepare students for entrance exams to France's prestigious higher education institutions.***Publications**

Article, Pepiot A., Rahib D., Velter A. and Breban R., *Epidemiological strategies based on self-screening tools: A modeling assessment*, under review, 2024. (see Chap. VII.3., pages 86-244 of the [thesis](#))

Article, Pepiot A., Supervie V. and Breban R. *Impact of voluntary testing on infectious disease epidemiology: A game theoretic approach*. PLOS ONE, 18(11): e0293968, November 2023 [[article](#), [preprint](#)]

Talks

Poster, Pepiot A., Supervie V. and Breban R. *Can self-testing end infectious disease epidemics? A tentative answer through game theory*, ECMTB, Heidelberg, September 2022 [[poster](#)]

Presentation, Pepiot A., Supervie V. and Breban R. *Can self-testing end infectious disease epidemics? A tentative answer through game theory*, Journée Scientifique Sidaction, Paris, March 2022

Presentation, Pepiot A., Supervie V. and Breban R. *Can self-testing end infectious disease epidemics? A tentative answer through game theory*, Journée des Jeunes Chercheurs Sidaction, online, November 2021

Poster, Pepiot A., Supervie V. and Breban R. *Impact de l'auto-dépistage sur l'épidémiologie des maladies infectieuses : Approche par la théorie des jeux*, Séminaire annuel de l'école doctorale Pierre Louis de santé publique, Saint Malo, October 2020 [[poster](#)]

Poster, Pepiot A., Supervie V. and Breban R. *Vers une élimination des maladies infectieuses avec l'auto-dépistage ? Approche par la théorie des jeux et application à l'épidémie du VIH*. Université des Jeunes Chercheurs Sidaction, Carry-le-Rouet, November 2019 [[poster](#)]

Skills

LANGUAGES

French - native**English** - proficient**German** - basic/intermediate

PROGRAMMING

Primary: Matlab, R, Python

Secondary: AMPL, Julia, C++, SAS

OTHERS

L^AT_EX, Microsoft Office, Git ([Github](#))